Experiment :-
Study of LAN Topology. Network devices and Address Configuration.

Objective :-

1) To Study the LAN topology used in College Campus
2) To identify networking device & understand their functions.
3) To analyze wired & wireless Communication method used in the network.
4) To measure Speed of network for wired & wireless Connections.
8) To obtained MAC, IP Address & Subnet mask

Problem Statement -1

Draw & identify LAN topology of College Campus & Clearly label all Components such as router, Switches & etc.

Soln :-

The College Campus network observed this Study Consists of:

* layer-3 Core Switches in the Server room.
* Managed by POE+ Switches.
* wireless Access point (APs)
* Ethernet port in Classroom & Laboratories.

Network Topology :-

(i) Core layer
(ii) Distribution layer
(iii) Access layer

These all are the network hierarchical layer.

Hybrid Hierarchical Topology which Combines.
(i) Tree topology
(ii) Extended Star topology
(iii) Ring Topology (Server room).

Core layer :-
- Core layer Contains Server Room and Server room have firewall, Router, 2- L3+ Switches patch Rack, Server, inverter etc.

Distribution layer :-
- Distribution layer Contains Server Room and
- The 4 Switches are-

(i) Main Build → R7 (Room no - 213, 2nd floor)
(ii) Main Build → R1 (Room no - 801, 3rd floor)
(iii) Main Build → R6 (Room no - 113, 1st floor)
(iv) Main Build → R2 (Room no - 201, 2nd floor)

- This 4 - Switches Contains →

(i) 18 port Gigabit Easy Smart Switch with 16 - port TL - SG1218.PE

(ii) SG 34 28XMP → 24 port Gigabit & 4 - port 10GE SFP+, L2+ Managed Switches with 24 port POE+

→ Now lets discuss there 2 above Components →

Now TP links TL - SR1218MPE
- Mainly used for : CCTV Cammera, wifi access point etc.
- 18 port Gigabit "Easy Smart" Switch
- 16 port Support POE+

(2) TP-Link Omada SG3428XMP

- Higher-end arranged Layer 2+ Switch
- 24 POE+ Port, uplinks, 10G SFP
- usually acts as a main distribution for each floor

Access Layer :-

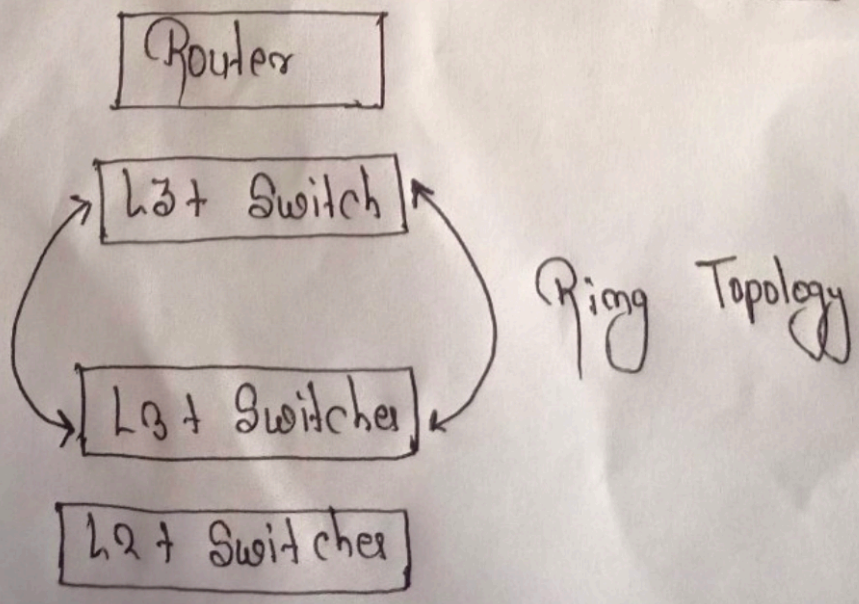
- Access layer Contain 3-Switches i.e

(i) Main Build R_1 - [Room no-301, 3rd floor]
(ii) Main Build R_4 - [Room no - Ground floor]
(iii) Main Build R_5 - [Room no - 18, Ground floor]

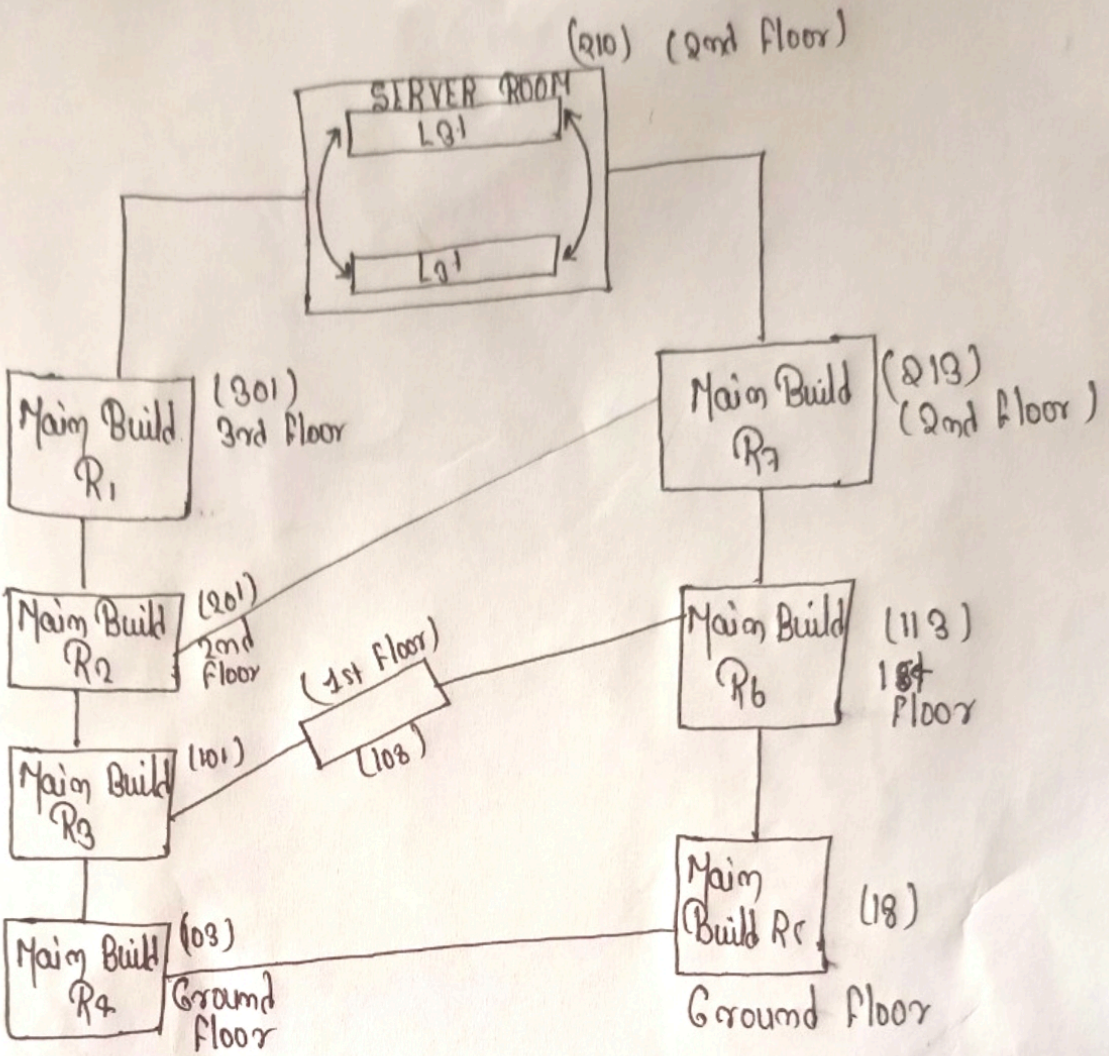
- Every Switches act as access layer wich helps to provide network Connectivity to all Class-room, Lab, faculty Room etc.
- Access Layer directly Connect :-

(i) Ethernet Port
(ii) Wireless AP
(iii) Lab System
(iv) Outdoor wireless AP (ex-018)

→ Server Room Architecture / Topology :-

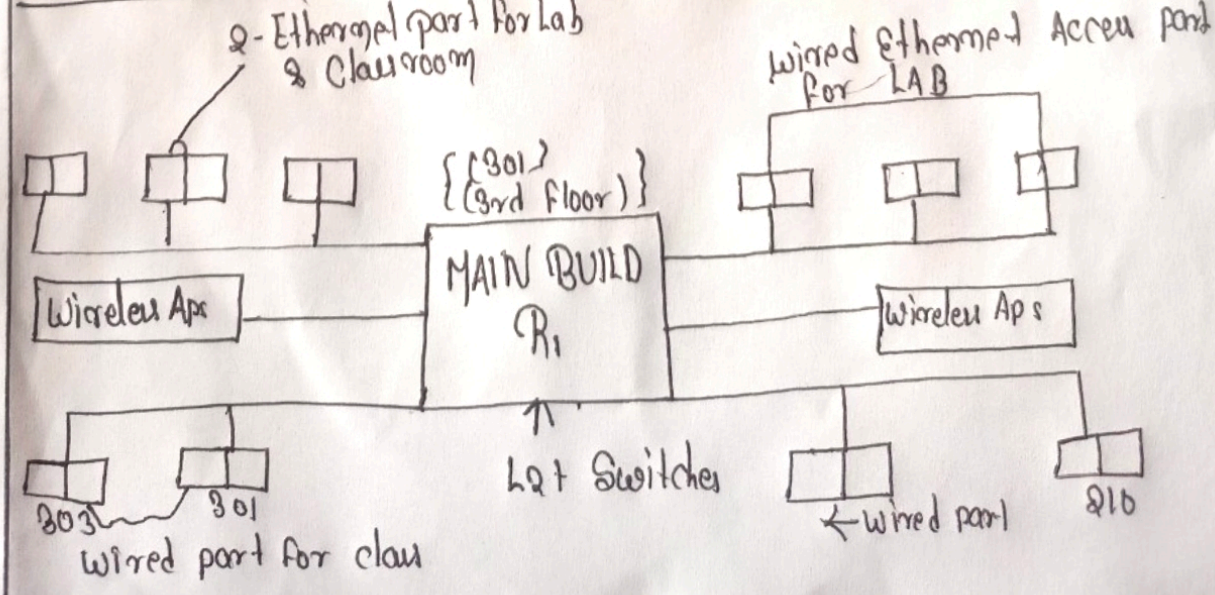


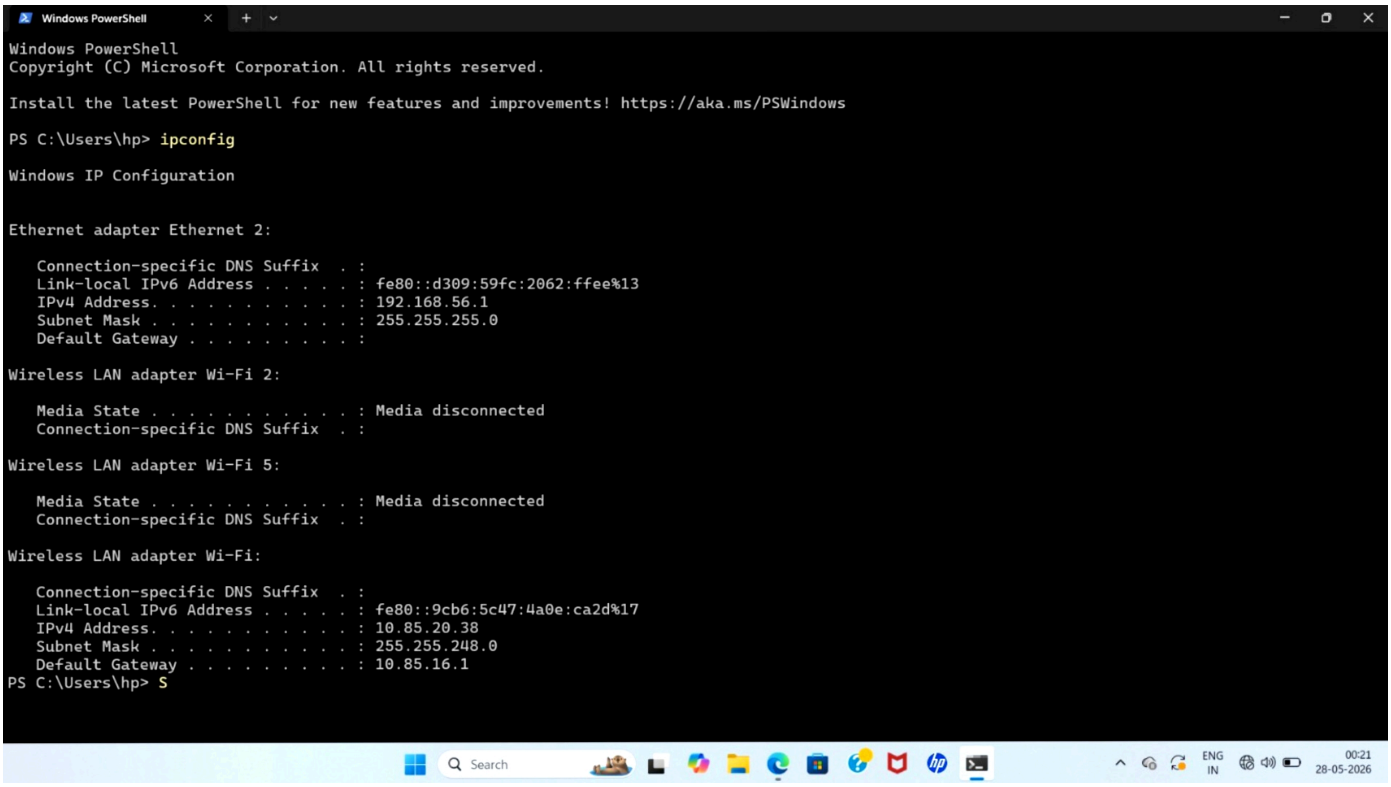
Ring Topology

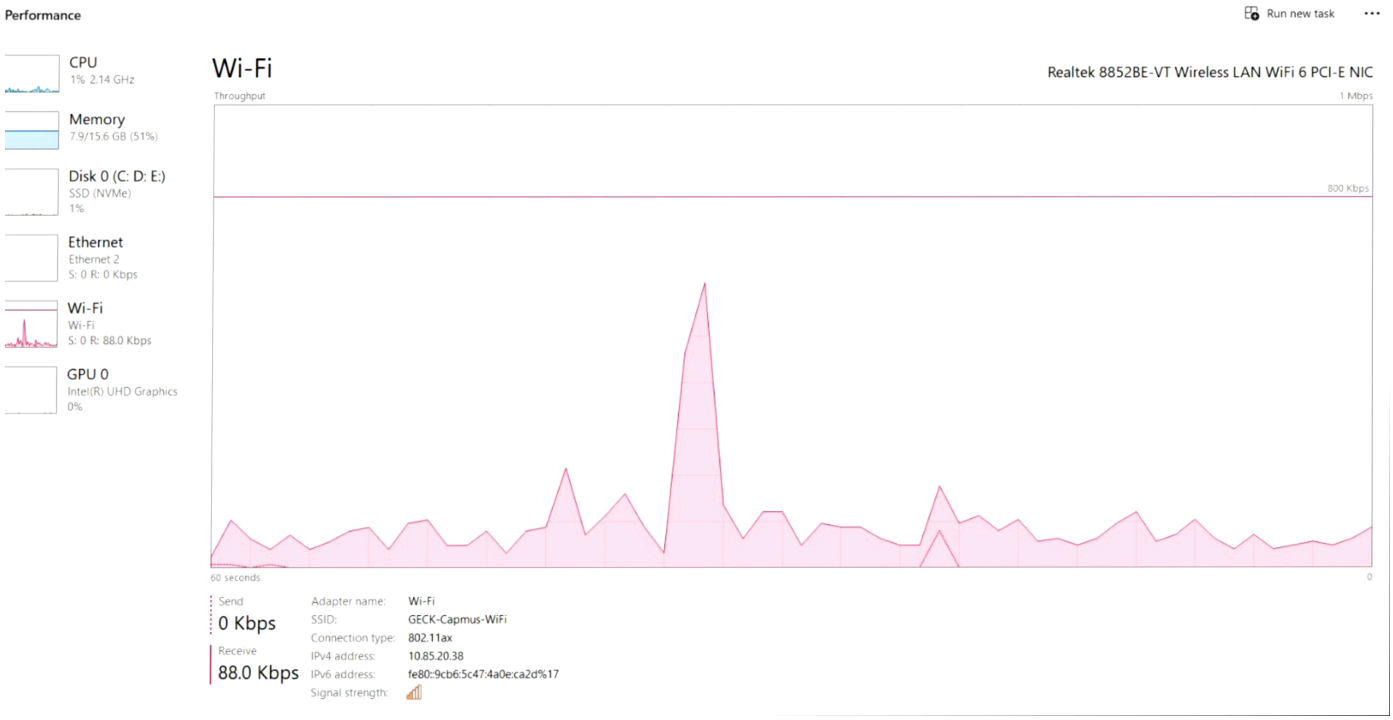


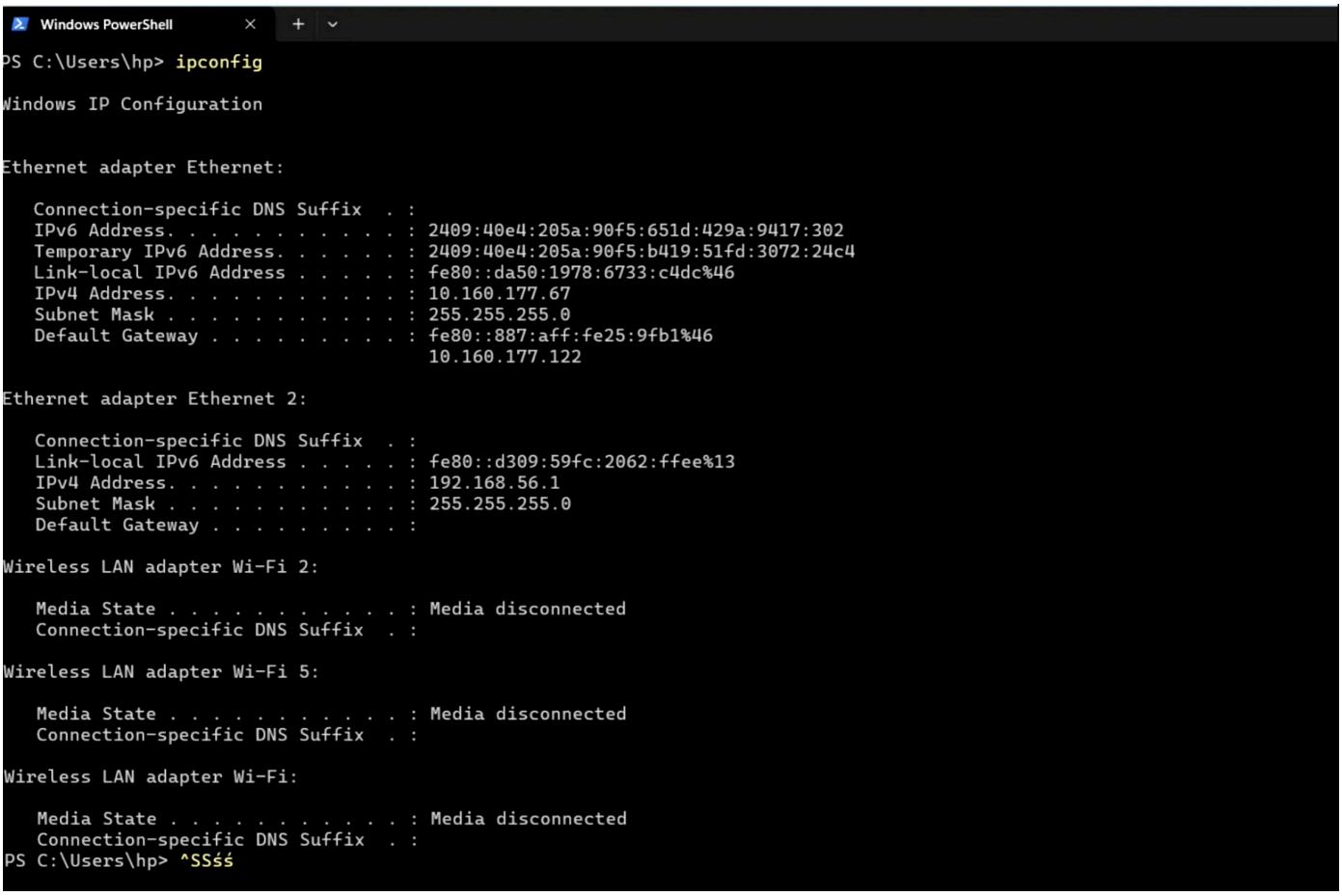
→ R1, R7, R2, R6 → Distribution layer
→ Server Room → Core layer
→ R1, R2, R3, R4, R5, R6, R7 → Access layer

ACCESS LAYER SWITCH (3rd Floor - 301)




Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\Users\hp> ipconfig
Windows IP Configuration
Ethernet adapter Ethernet 2:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::d309:59fc:2062:ffee%13
IPv4 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
Wireless LAN adapter Wi-Fi 2:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Wi-Fi 5:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::9cb6:5c47:4a0e:ca2d%17
IPv4 Address. . . . . : 10.85.20.38
Subnet Mask . . . . . : 255.255.248.0
Default Gateway . . . . . : 10.85.16.1
PS C:\Users\hp> S
Search
ENG
IN
00:21
28-05-2026


Performance
Run new task
CPU
1% 2.14 GHz
Memory
7.9/15.6 GB (51%)
Disk 0 (C: D: E:)
SSD (NVMe)
1%
Ethernet
Ethernet 2
S: 0 R: 0 Kbps
Wi-Fi
Wi-Fi
S: 0 R: 88.0 Kbps
GPU 0
Intel(R) UHD Graphics
0%
Wi-Fi
Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC
Throughput
1 Mbps
800 Kbps
60 seconds
0
Send
0 Kbps
Receive
88.0 Kbps
Adapter name: Wi-Fi
SSID: GECK-Capmus-WiFi
Connection type: 802.11ax
IPv4 address: 10.85.20.38
IPv6 address: fe80::9cb6:5c47:4a0e:ca2d%17
Signal strength:


Windows PowerShell
PS C:\Users\hp> ipconfig
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2409:40e4:205a:90f5:651d:429a:9417:302
Temporary IPv6 Address. . . . . : 2409:40e4:205a:90f5:b419:51fd:3072:24c4
Link-local IPv6 Address . . . . . : fe80::da50:1978:6733:c4dc%46
IPv4 Address. . . . . : 10.160.177.67
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::887:aff:fe25:9fb1%46
10.160.177.122
Ethernet adapter Ethernet 2:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::d309:59fc:2062:ffee%13
IPv4 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
Wireless LAN adapter Wi-Fi 2:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Wi-Fi 5:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Wi-Fi:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
PS C:\Users\hp> ^SSśś

→ Difference b/w wired & wireless Speed?

Wired Network	Wireless Network
• Faster Communication	• Slower Communication
• Stable Connection	• Affected by distance/ Interference
• Low latency	• Higher latency
• Dedicated physical medium	• Shared radio medium

→ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.

→ Wireless Connections uses radio waves which are affected by interference, distance, obstacles & Multiple Connected users.

8. Role of Subnet mask in network?

A Submask identifies the network portion & host portion of an Ip address. It help a device determine whether another device belongs to the same network or different network.

• It is important for:

(i) Network Segment

(ii) Efficient routing

(iii) Communication b/w devices.

OBSERVATION

During the Study of the College network infrastructure:-

(i) A Centralized Server room was used of network backbone.

(ii) Two L3 Switches were Connected in redundant Cyclic Structure.

(iii) Managed TP-link POE + Switches were used in network Cabinets.

Advantage of Existing Network?

- Scalable architecture
- Centralized Management
- Supports POE + device
- Easy expansion of floors & classroom
- Support both wireless & wired.
- Redundant link for failover.

Conclusion

The observed Campus LAN follow a Hybrid hierarchical topology Consist of tree, Extended Star, Ring. The network uses Manage of POE Switches wireless access point - Ethernet Port & Centralized Serverinfra Structure to provide efficient wired & wireless Communication, access the Campus.